

Relation between Class Groups of Binary Quadratic Forms and Ideal Class Groups in Quadratic Fields

Summer 2024

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Abstract

The first coherent exposition of the theory of binary quadratic forms was given by Gauss in the *Disquisitiones Arithmeticae*. During the nineteenth century, as the theory of ideals and the rudiments of algebraic number theory were developed, it became clear that this theory of binary quadratic forms, so elementary and computationally explicit, was indeed just a special case of a much more elegant and abstract theory which, unfortunately, is not computationally explicit. In recent years the original theory has been laid aside.

Although the theory of forms is just the quadratic case of results in algebraic number theory, the theorems can be independently proved using elementary methods. As Gauss recognized, it is truly delightful to build an elegant algebraic system from just the clever use of high school algebra.

The first goal of this presentation is to see the structure of ideals in Quadratic Fields following which we deduce a proposition which helps us to study a specific class of ideals thereby motivating us to study primitive forms and Dirichlet Composition which is key in giving the group structure to C_Δ i.e equivalence class of primitive binary quadratic forms.

Next we see some theory for general number rings i.e formation of class groups etc. At last we merge all of this together to derive two important results -

1. Giving group structure to C_Δ when Δ is field discriminant.
2. Proving finiteness of class number for Quadratic Fields.
(A special case of a more general question of "finiteness of class number in any Number Fields" in Algebraic Number Theory).

Notations

- $K \mapsto \mathbb{Q}(\sqrt{\Delta})$.
- $O_K \mapsto$ Ring of Integers of a Quadratic Number Field K .
- $\overline{\mathbb{Z}} \mapsto$ Ring of Algebraic Integers.
- $C_{O_K} \mapsto$ Ideal Class Group.
- $C_{O_K}^+ \mapsto$ Narrow Ideal Class Group.
- $O_K^* \mapsto$ Group of units in O_K .
- $N(\alpha) \mapsto$ Norm of α .
- $T(\alpha) \mapsto$ Trace of α .
- $Cl(\Delta) \mapsto$ set of all primitive forms with the given discriminant.
- $C_\Delta \mapsto$ the equivalence class of the primitive Binary Quadratic Forms under transformation by $SL_2(\mathbb{Z})$ i.e $Cl(\Delta)/\sim$.
- $\sim \mapsto$ Equivalence of Binary Quadratic Forms under $SL_2(\mathbb{Z})$ transformation.
- $I_A \mapsto$ set of all fractional A ideal is an Abelian group.
- $P_A \mapsto$ set of all principal fractional ideal is a subgroup of I_A .
- $P_A^+ \mapsto$ set of all principal fractional ideal is a subgroup of I_A with $N(\alpha) > 0$.
- $(a, b, c) \mapsto$ Primitive Binary Quadratic Form $ax^2 + bxy + cy^2$.
- $disc(B) \mapsto$ Discriminant of basis B of a Number Ring.

Contents

1	Algebra Preliminary	4
2	Just Enough Number Theory to Get Started	7
2.1	Number Fields	7
2.2	Quadratic Fields	7
2.3	Norm, Trace and Discriminant	9
3	Ideals in Quadratic Fields	11
4	Motivation for United Forms and Class Group	14
4.1	United Forms And Class Group	14
5	Fractional Ideals and Ideal Class Group	19
6	Main Results	22
6.1	$ C_{O_K} $ is finite for all Quadratic Number Rings	22
6.2	C_Δ has a group structure for all Fundamental Discriminants Δ	25

Chapter 1

Algebra Preliminary

Definition 1.0.1. Group is a pair $(G, \cdot)$ where G is a set and “ $\cdot$ ” is a binary operation for which the following holds.

1. G is closed under “ $\cdot$ ” i.e $(\cdot : G \times G \rightarrow G)$.
2. “ $\cdot$ ” is associative i.e $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
3. $\exists e \in G$ such that $a \cdot e = e \cdot a = a \quad \forall a \in G$.
4. For each $a \in G, \exists b \in G$ such that $a \cdot b = e = b \cdot a$.
 - If $a \cdot b = b \cdot a \quad \forall a, b \in G$, G is called commutative (Abelian Group).
 - Order of a group is cardinality of G .

Definition 1.0.2. If $H \subset G$ such that H is a group in itself then H is called a subgroup of G .

Theorem 1.0.1. (*Lagrange Theorem*): $[G : H] = \frac{|G|}{|H|}$.

Definition 1.0.3. In our case every ring we talk about would be commutative and would have a unity denoted as 1_R .

Ring is a triple $(R, +, \cdot)$ where R is a set equipped with two binary operations on R for which

1. $(R, +)$ forms an Abelian Group.
2. R is closed under “ $\cdot$ ”
3. “ $\cdot$ ” is associative.
4. $a \cdot 1_R = 1_R \cdot a = a \quad \forall a \in R$.
5. $a \cdot b = b \cdot a \rightarrow$ Since R is a commutative ring.

Also we have $a \cdot (b + c) = a \cdot b + a \cdot c$ [“.” distributes over “+”].

Definition 1.0.4. Let R be a ring and $I \subseteq R$ such that

1. $\forall a, b \in I, a + b \in I$.
2. If $a \in I$, then $\forall r \in R, ra \in I$.

then I is called an ideal of R .

Theorem 1.0.2. (*Chinese Remainder Theorem*): If A & B are 2 coprime ideals i.e $A + B = R$ then $A \cap B = AB$.

$$R/AB = R/A \cap B \cong R/A \times R/B.$$

Definition 1.0.5. Let A be a ring and M be an Abelian group together with an operation

$$\mu : A \times M \rightarrow M$$

$$\mu(a, m) \mapsto am$$

such that following is satisfied

1. $a(x + y) = ax + ay$
2. $(a + b)x = ax + bx$.
3. $(ab)x = a(bx)$.
4. $1x = x$

The pair (M, μ) is called a A -Module.

- R/I is an integral domain if and only if I is prime ideal.
- R/I is a field if and only if I is maximal ideal.

Definition 1.0.6. If in a ring, the elements in $(R \setminus \{0\})$ are invertible then R is called a field.

Let L & K be two fields if $\exists$ injective homomorphism from L to K . i.e. $\sigma : L \rightarrow K$ such that

$$\sigma(a + b) = \sigma(a) + \sigma(b)$$

$$\sigma(ab) = \sigma(a)\sigma(b)$$

$$\sigma(1_L) = 1_K.$$

& σ is injective, then K is called a field extension of L & $[L : K]$ is the degree of field extension i.e number of basis elements needed when L is looked as vector space over K .

In our case all degree of extensions would be finite i.e the the basis would be finite.

If $M \subset N \subset L$ are fields then

$$[L : M] = [L : N][N : M]$$

Theorem 1.0.3. (*Primitive Element Theorem*): If $[L : K] = n$ and $L = K(\alpha_1, \alpha_2, \dots, \alpha_n)$ then each α_i is algebraic over K and $L = K(\alpha)$ for some $\alpha \in L$.
 $K[\alpha_1, \dots, \alpha_n] = K(\alpha_1, \dots, \alpha_n)$ if all α_i s are algebraic over K .

Let

$$\begin{aligned}\varphi : F &\longmapsto F' \\ \varphi(a) &= a'\end{aligned}$$

be a (field) isomorphism, and let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

be an arbitrary polynomial in $F[x]$. We extend φ to a ring isomorphism $\bar{\varphi}$ by defining

$$\begin{aligned}\bar{\varphi} : F[x] &\longmapsto F'[x] \\ \bar{\varphi}(f(x)) &= f'(x)\end{aligned}$$

where

$$f'(x) = a'_n x^n + a'_{n-1} x^{n-1} + \dots + a'_1 x + a'_0$$

Then we have-

Theorem 1.0.4. (*Isomorphism Extension Theorem*): Let $\varphi : F \longmapsto F'$ be an isomorphism, let $f(x)$ be a polynomial in $F[x]$ that is irreducible over F , and let $f'(x)$ be as defined as above. Let α be a root of $f(x)$ and let α' be a root of $f'(x)$. Then φ extends uniquely to an isomorphism $\tau : F(\alpha) \longmapsto F'(\alpha')$ such that $\tau(\alpha) = \alpha'$ and $\tau|_F = \varphi$.

$$\begin{array}{ccc} F & \xrightarrow{\varphi} & F' \\ \downarrow & & \downarrow \\ F(\alpha) & \xrightarrow{\tau} & F'(\alpha') \end{array}$$

Chapter 2

Just Enough Number Theory to Get Started

2.1 Number Fields

A finite field extension of $\mathbb{Q}$ is called a Number field. if L is a Number field, $[L : \mathbb{Q}] < \infty$ (say n).

Hence

$$L = \mathbb{Q} [\alpha_1, \dots, \alpha_n] = \mathbb{Q}(\alpha_1, \dots, \alpha_n).$$

Since $[L : \mathbb{Q}] < \infty$ we have α_i algebraic over $\mathbb{Q}$ by [Theorem 1.0.3](#), we have

$$L = \mathbb{Q}(\alpha_1, \dots, \alpha_n) = \mathbb{Q}(\alpha) \text{ for some } \alpha \in L.$$

Hence any number field is of the $\mathbb{Q}(\alpha)$.

If $[L : \mathbb{Q}] = 2$ we call L a quadratic number field.

2.2 Quadratic Fields

Proposition 2.2.1. *L is a quadratic number field if and only if $L = \mathbb{Q}(\sqrt{m})$, $m \in \mathbb{Z}$ which is not a perfect square.*

Proof. If $L = \mathbb{Q}(\sqrt{m})$, $m \in \mathbb{Z}$,
then $\{1, \sqrt{m}\}$ are $\mathbb{Q}$ -linear-independent.

$$\begin{aligned} & \because q_1 + q_2\sqrt{m} = 0 \\ & \Rightarrow q_1^2 = mq_2^2 \\ & \Rightarrow a_1^2 = a_2^2m \text{ [after clearing denominator]}. \end{aligned}$$

Contradiction [$\because m$ is not a perfect square]. Thus $[L : \mathbb{Q}] = 2$ and L is a Quadratic Field.

Conversely if $[L : \mathbb{Q}] = 2$ clearly $\exists \alpha \in L \setminus \mathbb{Q}$.

$\Rightarrow a\alpha^2 + ba + c = 0; a, b, c \in \mathbb{Z}$
 $\Rightarrow \alpha = \frac{-b \pm \sqrt{b^2 - 4ca}}{2a}$, clearly $b^2 - 4ca$ is not a perfect square as $\alpha \notin \mathbb{Q}$.
 Let $m = b^2 - 4ac$.

Claim 2.2.1. $L = \mathbb{Q}(\sqrt{m})$.

Proof. Clearly $\mathbb{Q}(\sqrt{m}) \subseteq L$
 if $\mathbb{Q}(\sqrt{m}) \neq L$ $[\because \mathbb{Q} \subseteq \mathbb{Q}(\sqrt{m}) \subseteq L]$. We can say

$$\begin{aligned}
 [L : \mathbb{Q}] &= [L : \mathbb{Q}(\sqrt{m})][\mathbb{Q}(\sqrt{m}) : \mathbb{Q}] \\
 \Rightarrow [L : \mathbb{Q}(\sqrt{m})] &= 1 \quad \Rightarrow L = \mathbb{Q}(\sqrt{m}). \quad [\because [\mathbb{Q}(\sqrt{m}) : \mathbb{Q}] = 2]
 \end{aligned}$$

□

This completes our proof. □

Hence from now on we denote a quadratic number field as K or $\mathbb{Q}(\sqrt{d})$ for some $d \in \mathbb{Z}$ & d not a perfect square.

Definition 2.2.1. $O_K = \overline{\mathbb{Z}} \cap K$. [i.e $\alpha \in K$ such that $f(\alpha) = 0$; f is monic & $f \in \mathbb{Z}[x]$].

[i.e seeing the ring O_K is the integral closure of $\mathbb{Z}$ in K].

Proposition 2.2.2. $O_K = \begin{cases} \mathbb{Z}[\sqrt{d}], & d \equiv 2, 3 \pmod{4}. \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right], & d \equiv 1 \pmod{4}. \end{cases}$

Proof. Every element $\alpha \in \mathbb{Q}(\sqrt{d})$ is of the form $\alpha = r + s\sqrt{d}$. for $r, s \in \mathbb{Q}$. Hence

$$\alpha = \frac{a + b\sqrt{d}}{c}$$

Where $a, b, c \in \mathbb{Z}, c > 0$ and no prime divides all of a, b, c . Now α is an algebraic integer if the coefficients of the minimal polynomial.

$$\left(t - \left(\frac{a + b\sqrt{d}}{c}\right)\right) \left(t - \left(\frac{a - b\sqrt{d}}{c}\right)\right) \text{ are integers.}$$

Thus

$$\frac{a^2 - b^2d}{c^2} \in \mathbb{Z} \dots\dots(1) \quad \& \quad \frac{2a}{c} \in \mathbb{Z} \dots\dots(2)$$

if c & a have common factor p then (1) implies that $p \mid b$ because (d is square free), which contradicts that a, b, c do not have common factor.

$\Rightarrow c \mid 2 \Rightarrow c = 1$ or $c = 2$

if $c = 1$ then α is an integer of K in any case, so we may concentrate on $c = 2$.

Now a & b both must be odd & $\frac{a^2 - b^2d}{4} \in \mathbb{Z}$

Hence $a^2 - b^2d \equiv 0 \pmod{4}$. $a^2 = b^2d \equiv 1 \pmod{4} \Rightarrow d \equiv 1 \pmod{4}$.

Conversely if $d \equiv 1 \pmod{4}$ then odd a & b we have α an integer because (1) & (2) holds.

This completes the proof. □

Proposition 2.2.3. Let $K = \mathbb{Q}(\theta)$ be a number field of degree n over $\mathbb{Q}$. Then there are exactly n distinct monomorphism $\sigma_i : K \rightarrow \mathbb{C}$ $\{i \in \{1, \dots, n\}\}$. The elements $\sigma_i(\theta) = \theta_i$ are distinct zeros in $\mathbb{C}$ of $f(x)$ (minimum polynomial of θ over $\mathbb{C}$).

Proof. Let $\{\theta_1, \theta_2, \dots, \theta_n$ be distinct roots of $f(x)$ (minimum polynomial of θ over $\mathbb{C}$).

If α is a root of $f(x)$.

$\Rightarrow \sigma(\alpha)$ is also a root of f .

(Just apply σ to $(\alpha^n + \sum_{i=0}^{n-1} q_i \alpha^i) = 0$)

$$\sigma \left(\alpha^n + \sum_{i=0}^{n-1} q_i \alpha^i \right) = 0.$$

$$\because \sigma(q) = q \quad \forall q \in \mathbb{Q}.$$

Let $S_1 = \{ \text{set of all distinct monomorphism from } K \rightarrow \mathbb{C} \}$ and $S_2 = \{ \text{set of all distinct roots (has to be } \because f \text{ is irreducible) of } f \}$.

If $\sigma \in S_1 \Rightarrow \sigma \in S_2$

$$\Rightarrow S_1 \subseteq S_2 \Rightarrow |S_1| \leq |S_2|$$

Now from [Theorem 1.0.4](#) if

$\varphi : \mathbb{Q} \rightarrow \mathbb{Q}$ be the identify map.

Let f be the irreducible polynomial over $\mathbb{Q}$ with a root θ . Let θ_i , be a other root of f .

Then $\exists! \tau : \mathbb{Q}(\theta) \rightarrow \mathbb{Q}(\theta_i)$ such that $\tau|_{\mathbb{Q}} = \varphi$, & $\tau(\theta) = \theta_i$.

$$\begin{array}{ccc} \mathbb{Q} & \xrightarrow{\varphi} & \mathbb{Q} \\ \downarrow & & \downarrow \\ \mathbb{Q}(\theta) & \xrightarrow{\tau} & \mathbb{Q}(\theta_i) \end{array}$$

Each such τ is also a monomorphism from $K \rightarrow \mathbb{C}$. So for each $\theta_i \in S_2$, we can associate a unique monomorphism ie an element of S_1 , $\Rightarrow |S_2| \leq |S_1|$. Together with $|S_1| \leq |S_2|$ we get $|S_1| = |S_2|$ & $|S_2| = |S_1|$, which completes our proof. $\square$

If $K = \mathbb{Q}(\sqrt{d})$ we have 2 such monomorphism.

$$\sigma_1(\sqrt{d}) = \sqrt{d}, \quad \sigma_2(\sqrt{d}) = -\sqrt{d}.$$

Thus $\sigma_i(\alpha + \beta\sqrt{d}) = \alpha + \beta\sigma_i(\sqrt{d})$ ($\because \sigma_i(q) = q \quad \forall q \in \mathbb{Q}$)

2.3 Norm, Trace and Discriminant

Definition 2.3.1. Let $\alpha \in K$

$$\text{Norm}(\alpha) = \sigma_1(\alpha)\sigma_2(\alpha)$$

$$\text{Trace}(\alpha) = \sigma_1(\alpha) + \sigma_2(\alpha).$$

Let $\{1, \sqrt{d}\}$ be the basis of $\mathbb{Z}[\sqrt{d}]$ over $\mathbb{Z}$.

Definition 2.3.2. Discriminant (denoted by D or $\text{Disc}(B)$) of $B = \{1, \sqrt{d}\}$ is defined as

$$D = \left| \begin{pmatrix} \sigma_1(1) & \sigma_1(\sqrt{d}) \\ \sigma_2(1) & \sigma_2(\sqrt{d}) \end{pmatrix} \right|^2 \\ = 4d$$

& $B = \left\{1, \frac{1+\sqrt{d}}{2}\right\}$ to be,

$$D = \left(\det \begin{pmatrix} \sigma_1(1) & \sigma_1\left(\frac{1+\sqrt{d}}{2}\right) \\ \sigma_2(1) & \sigma_2\left(\frac{1+\sqrt{d}}{2}\right) \end{pmatrix} \right)^2 \\ = d.$$

By a general result about number ring we can show $\text{Disc}(B)$ is an invariant for any basis of O_K . So we may choose here $\{1, \sqrt{d}\}$ or $\left\{1, \frac{1+\sqrt{d}}{2}\right\}$ to work with.

$$\text{disc}(B) = \begin{cases} 4d, & \text{if } d \equiv 2, 3 \pmod{4} \\ d, & \text{if } d \equiv 1 \pmod{4}. \end{cases}$$

Proposition 2.3.1. u is a unit in O_K if and only if $\text{Norm}(x) = \pm 1$.

For the sake of simplicity we henceforth write Norm with N and Trace as T .

Proof. If x is a unit $\Rightarrow xu = 1 \Rightarrow N(x)N(u) = 1 \Rightarrow N(x) \in \mathbb{Z}$ & $N(u) = \pm 1$.
Conversely if $N(u) = \pm 1$.

$$\Rightarrow \prod \sigma_i(u) = \pm 1 \quad \text{Let } \sigma_1(x) = x, \quad x \in k.$$

Thus take $x = \sigma_1(x)$ & u to be $\prod_{i \geq 2} \sigma_i(x)$. This completes our proof. □

Proposition 2.3.2. Let O_K^* denote the group of units in O_K .

$$O_K^* = \begin{cases} \mathbb{Z} \times \langle -1 \rangle, & d > 1 \\ \langle i \rangle, & d = -1. \\ \left\langle \frac{1+\sqrt{3}i}{2} \right\rangle, & d = -3, \\ \langle -1 \rangle, & d = -2 \text{ \& } d < -3 \end{cases}$$

Proof. Check the Norms for $d \leq -1$,
 $d > 1$ case can be done using Pell's equation. □

Chapter 3

Ideals in Quadratic Fields

We would study structure of ideals in $\mathbb{Q}(\sqrt{d})$.
Throughout this section we will write

$$\omega = \begin{cases} \sqrt{d}, & d \equiv 2, 3 \pmod{4} \\ \frac{1+\sqrt{d}}{2}, & d \equiv 1 \pmod{4} \end{cases}$$

$$\omega' = \begin{cases} -\sqrt{d}, & d \equiv 2, 3 \pmod{4} \\ \frac{1-\sqrt{d}}{2}, & d \equiv 1 \pmod{4} \end{cases}$$

Proposition 3.0.1. : Every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$ for some $m \geq 0$.

Proof. Take $a \in H$. such that $|a| \geq 0$ & $|a|$ is least. If $|a|$ is 0 then done, if not then

Claim 3.0.1. : $H = a\mathbb{Z}$.

$\because a \in H \quad a\mathbb{Z} \subseteq H$.

Now let $x \in H$, if $x \notin a\mathbb{Z} \Rightarrow \exists y$ and z such that

$$\underbrace{x}_{\in H} = \underbrace{ay}_{\in H} + z \Rightarrow z \in H$$

but $|z| < |a|$, a contradiction. □

Proposition 3.0.2. Let $M \subset O_K$ be a $\mathbb{Z}$ -module in O_K . Then $\exists m$ & $n \in \mathbb{N}$ and $a \in \mathbb{Z}$ such that $M = n\mathbb{Z} \oplus (a + m\omega)\mathbb{Z}$.

Proof. Consider the subgroup of $\mathbb{Z}$

$$H = \{s \in \mathbb{Z} : \exists r \in \mathbb{Z} \text{ for which } r + s\omega \in M\}$$

Every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$, for some $m \geq 0 \Rightarrow m \in H$. Hence $\exists a \in \mathbb{Z}$ such that $a + m\omega \in M$. Finally $M \cap \mathbb{Z}$ is also a subgroup of $\mathbb{Z} \Rightarrow M \cap \mathbb{Z} = n\mathbb{Z}$ for some $n \geq 0 \Rightarrow n \in M \cap \mathbb{Z}$.

Claim 3.0.2. $n\mathbb{Z} \cap (a + m\omega)\mathbb{Z} = \{0\}$ for $m, n \in \mathbb{N}$ and $a \in \mathbb{Z}$.

Proof. Let $x \in n\mathbb{Z} \cap (a + m\omega)\mathbb{Z}$ and $x \neq 0$.

$\Rightarrow x = nc$ for some $c \in \mathbb{Z} \setminus \{0\}$ and $x = (a + m\omega)b$ for some $b \in \mathbb{Z} \setminus \{0\}$.

$\Rightarrow nc = ab + mb\omega$

$\Rightarrow mb\omega \in \mathbb{Z}$

$\Rightarrow \omega \in \mathbb{Q}$, which is a contradiction since we assumed that d is not a perfect square and hence $\sqrt{d} \notin \mathbb{Q}$. This completes our claim. $\square$

Claim 3.0.3. $M = n\mathbb{Z} \oplus (a + m\omega)\mathbb{Z}$

$\because (a + m\omega) \in M \Rightarrow (a + m\omega)\mathbb{Z} \in M$.

$\& \therefore n\mathbb{Z} \in M \Rightarrow n\mathbb{Z} \oplus (a + m\omega)\mathbb{Z} \subseteq M$.

Now let $r + s\omega \in M$. $\because s \in H \Rightarrow s = mu$ for some $u \in \mathbb{Z}$.

$$\underbrace{r - ua}_{\in \mathbb{Z}} = \underbrace{r + s\omega}_{\in M} - \underbrace{u(a + m\omega)}_{\in M} \in M \cap \mathbb{Z}$$

$\Rightarrow r - ua = vn$ for some $v \in \mathbb{Z}$

$\Rightarrow r + s\omega = r - ua + ua + vn = vn + u(a + m\omega) \in n\mathbb{Z} \oplus (a + m\omega)\mathbb{Z}$

$\Rightarrow M \subseteq n\mathbb{Z} \oplus (a + m\omega)\mathbb{Z}$.

This completes our proof. $\square$

Our proof also showed that m & n are indeed unique.

Proposition 3.0.3. A non-zero $\mathbb{Z}$ -module $M = [n, a + m\omega]$ is an ideal if and only if $m \mid n, m \mid a$ i.e $a = mb$ & $n \mid m \cdot N(b + \omega)$.

Proof. Since M is an ideal (O_K -ideal)

$c \in M$ & $\omega \in O_K \Rightarrow c\omega \in M$.

$\Rightarrow c \in \mathbb{Z} \Rightarrow c\omega \in M$. [$\because \omega \in O_K$]

$\Rightarrow c \in H \Rightarrow m \cap \mathbb{Z} \subseteq H$.

$\Rightarrow n\mathbb{Z} \subseteq m\mathbb{Z} \Rightarrow m \mid n$. [assuming $c \in M \cap \mathbb{Z} \Rightarrow c \in H$].

To show $m \mid a$, we see that $w^2 \in O_K$.

$\Rightarrow \omega^2 = x + y\omega$ for some $x, y \in \mathbb{Z}$ [$\because M$ is an ideal]

$a + m\omega \in M \Rightarrow (a + m\omega)\omega = mx + (a + my)\omega \in M$.

$\Rightarrow a + my \in H \Rightarrow a + my$ is a multiple of $m \Rightarrow m \mid a \Rightarrow a = mb$.

Let $\alpha = a + m\omega = m(b + \omega)$, Then $\alpha \in M \Rightarrow \alpha(b + \omega') \in M$ [$\because M$ is an ideal.]

$\Rightarrow N(\alpha) = m^2(b + \omega)(b + \omega')$

$\Rightarrow \frac{N(\alpha)}{m} = m(b + \omega)(b + \omega') = \alpha(b + \omega') \in M$

$\because N(b + \omega) = (b + \omega)(b + \omega') \in \mathbb{Z} \Rightarrow mN(b + \omega) \in \mathbb{Z} \Rightarrow mN(b + \omega) \in M \cap \mathbb{Z}$

$\Rightarrow n \mid m \cdot N(b + \omega)$.

For the other direction it is easy to verify that if these conditions are met then it is indeed an ideal. $\square$

Hence we denote.

$$M = \langle n, a + m\omega \rangle.$$

Now that we have seen structure of ideals in Quadratic Fields we get back to work by looking at a special kind of ideal which would be useful from now on.

Chapter 4

Motivation for United Forms and Class Group

4.1 United Forms And Class Group

Throughout this section we will write-

$$\begin{aligned}\Delta &= \begin{cases} 4d, & \text{if } d \equiv 2, 3 \pmod{4} \\ d, & \text{if } d \equiv 1 \pmod{4}. \end{cases} \\ d &= \begin{cases} \frac{\Delta}{4}, & \text{if } d \equiv 2, 3 \pmod{4} \\ \Delta, & \text{if } d \equiv 1 \pmod{4}. \end{cases} \\ O_K &= \begin{cases} \mathbb{Z}[\frac{\sqrt{\Delta}}{2}], & \frac{\Delta}{4} \equiv 2, 3 \pmod{4} \\ \mathbb{Z}[\frac{1+\sqrt{\Delta}}{2}], & \Delta \equiv 1 \pmod{4} \end{cases} \\ \omega &= \begin{cases} \frac{\sqrt{\Delta}}{2}, & \frac{\Delta}{4} \equiv 2, 3 \pmod{4} \\ \frac{1+\sqrt{\Delta}}{2}, & \Delta \equiv 1 \pmod{4} \end{cases}\end{aligned}$$

Proposition 4.1.1. *If $a_1 \mid \frac{b_1^2 - \Delta}{4}$ & $a_1 > 0$ then $I_1 = \langle a_1, \frac{b_1 + \sqrt{\Delta}}{4} \rangle$ is an O_K -ideal.*

Proof. If b_1 is even $\Rightarrow \Delta$ is even.

Therefore $O_K = \mathbb{Z} \left[\frac{\sqrt{\Delta}}{2} \right]$ & $w = \frac{\sqrt{\Delta}}{2}$.

To show $I_1 = [n, a + m\omega]$ for some $m, n \in \mathbb{N}$ & $a \in \mathbb{Z}$.

Choose $n = a_1, a = \frac{b_1}{2}, m = 1$ and it works also clearly $m \mid n$ & $m \mid a$ & $a_1 \mid \left(\frac{b_1}{2} + \omega \right) \left(\frac{b_1}{2} + \omega' \right)$. where ω' is defined as in previous section.

$$\Rightarrow a_1 \mid \frac{b_1^2 - \Delta}{4}$$

Hence I_1 is an O_K -ideal.

If b_1 is odd $\Rightarrow \Delta$ is odd.

$$O_K = \mathbb{Z} \left[\frac{1 + \sqrt{\Delta}}{2} \right] \quad \omega = \frac{1 + \sqrt{\Delta}}{2}.$$

To show

$$I_1 = \left[n, a + m \frac{(1 + \sqrt{\Delta})}{2} \right]$$

Again choose $m = 1, a = \frac{b_1-1}{2}, n = a_1$, and check all the conditions are indeed satisfied.

Hence in either case. $I_1 = \left[a, \frac{b_1 + \sqrt{\Delta}}{2} \right]$ is an O_K ideal with

$n = a_1, a = \frac{b_1}{2}, m = 1$ if $b_1 = \text{even}$

$n = a_1, a = \frac{b_1-1}{2}, m = 1$ if $b_1 = \text{odd}$. □

From our condition $a_1 \mid \frac{b_1^2 - \Delta}{4}$ its natural to think of binary quadratic forms & so we do, from now on- set $K = \mathbb{Q}(\sqrt{\Delta})$ and set $Disc = \text{disc} \left\{ 1, \frac{\sqrt{\Delta}}{2} \right\}$ or $\text{disc} \left\{ 1, \frac{1 + \sqrt{\Delta}}{2} \right\}$ accordingly, where Δ is the discriminant associated with the binary quadratic form $f = (a_1, b_1, c_1)$. (primitive i.e $\gcd(a_1, b_1, c_1) = 1$).

- It is required to note that from now onwards we would only consider those Binary Quadratic Forms whose discriminants (Δ) is field discriminant for some Quadratic Field $\mathbb{Q}(\sqrt{\Delta_K})$ ($\Delta = \Delta_K$) or in other words satisfy the conditions:
 $\frac{\Delta}{4} \equiv 2, 3 \pmod{4}$ if $\Delta \equiv 0 \pmod{4}$ or
 $\Delta \equiv 1 \pmod{4}$
and the binary quadratic form should be primitive.
- (i.e we wont be considering primitive forms with discriminant say 16 for eg. (3,32,64) won't be considered or if $\Delta \equiv 0 \pmod{4}$ we would be considering primitive forms with discriminant of the form $16k + 8$ or $16k + 12$ and of the form $4k + 1$ if $\Delta \equiv 1 \pmod{4}$).
- We call Δ (discriminant of a binary quadratic form) fundamental discriminant if ($\Delta = \Delta_K$) for some Quadratic Field $\mathbb{Q}(\sqrt{\Delta_K})$.
- Thereby we try to give a group structure to C_Δ for all Δ of the form $4k+1, 16k+8, 16k+12$ under Dirichlet composition leaving out on all Δ of the form $16k+4$ and $16k$
- For each form (a, b, c) associate the ideal
 $\left\langle a, \frac{-b + \sqrt{\Delta}}{2} \right\rangle \rightarrow$ (it is indeed an ideal since it satisfies our previous requirements).

If

$$I_1 = \left\langle a_1, \frac{b_1 + \sqrt{\Delta}}{2} \right\rangle$$

$$I_2 = \left\langle a_2, \frac{b_2 + \sqrt{\Delta}}{2} \right\rangle$$

Then check-

$$I_1 I_2 = (g) \left\langle a_3, \frac{b_3 + \sqrt{\Delta}}{2} \right\rangle$$

Where

$$\begin{aligned} a_3 &= \frac{a_1 a_2}{g^2} \\ g &= \gcd \left(a_1, a_2, \frac{b_1 + b_2}{2} \right) \\ b_3 &= \frac{1}{g} \left(\delta a_2 b_1, \mu a_1 b_2, \frac{\nu}{2} (b_1 b_2 + \Delta) \right) \pmod{2a_3} \\ \delta a_2 + \mu a_1 + \frac{\nu}{2} (b_1 + b_2) &= g. \end{aligned}$$

If g was “1” it would have been great since we could get back an ideal. of the same form & would then correspond to (a_1, b_3, c_1) .

Thus

$$\begin{aligned} a_3 &= a_1 a_2. \\ \gcd \left(a_1, a_2, \frac{b_1 + b_2}{2} \right) &= 1. \\ \& \ b_3 &\equiv \left(\delta a_2 b_1 + \mu a_1 b_2 + \frac{\nu}{2} (b_1 b_2 + \Delta) \right) \pmod{2a_3}. \end{aligned}$$

Where δ, μ, ν all are integers.

Indeed such b_3 is unique satisfying congruences-

$$b_3 \equiv b_1 \pmod{2a_1}$$

and

$$b_3 \equiv b_2 \pmod{2a_2}$$

This gave the concept of united forms i.e:-

$f=(a_1, b_1, c_1)$ & $g=(a_2, b_2, c_2)$ are united if $\gcd(a_1, a_2, \frac{(b_1+b_2)}{2}) = 1$.

Then $a_3 = a_1 a_2, b_3$ is the unique integer satisfying above congruence &

$$c_3 = \frac{b_3^2 - \Delta}{4a_3}.$$

Now define Dirichlet composition of f & g to be the form.

$$f \circ g = (a_3, b_3, c_3).$$

So far we have only defined Dirichlet composition for united form.

Another thing we must notice is that if

$$(ac, b, 1) \sim \begin{pmatrix} \frac{b}{2} & -1 \\ -1 & 0 \end{pmatrix} (1, 0, \Delta/4). \text{ When } \Delta \equiv 0 \pmod{4}.$$

$$(ac, b, 1) \sim \begin{pmatrix} \frac{-(b+1)}{2} & -1 \\ -1 & 0 \end{pmatrix} (1, 0, (1 - \Delta)/4). \text{ When } \Delta \equiv 1 \pmod{4}.$$

and also if $f = (a, b, c)$

$(a, b, c) \circ (1, 0, \Delta/4) = (a, b, c)$ and respectively $(1, 1, (1 - \Delta)/4)$ [$\because f$ is positive definite].

$$\because |b| < |a|.$$

$$B \equiv b \pmod{a}.$$

Hence $B = b$.

and

$$(a, b, c) \circ (c, b, a) = (ac, b, 1) \sim (1, 0, \Delta/4) \text{ or } (1, 1, (1 - \Delta)/4).$$

All the forms that we have done or going to do would be primitive forms.

Remark 4.1.1. Its required to note that we have only associated ideals to forms for with $a > 0$ i.e positive definite binary quadratic forms. But for indefinite forms there may be the case that $a < 0$. But we can sidestep that using a clever trick that we will see later.

It would have been great if C_Δ (the equivalence class of the primitive binary quadratic forms) could form a group but that's too early to say as we have just defined Dirichlet composition for united forms only and not general two primitive forms but nevertheless we would see that it is indeed true that C_Δ forms a group.

Indeed if we use united forms then from our previous calculations we saw that composition of forms $\xrightarrow{\text{corresponds to}}$ multiplication of ideals. Thus atleast for united forms Dirichlet composition induces a well defined binary composition.

As we would see in next section that C_{O_K} forms a group and we know that the cardinality of C_Δ is finite [NZM]. So if we can somehow show a bijection between them then we would prove that C_{O_K} is finite and C_Δ forms a group under Dirichlet composition. This is exactly what we would try to do from now on.

Remark 4.1.2. A small thing to note is that according to our restrictions mentioned in the beginning of the section we see that we can prove finiteness of class group for all quadratic number rings since given any discriminant with those conditions we can always find some primitive form with that given discriminant since we have to solve $b^2 - 4ac = \Delta$ in integers. So we can choose $a = 1$ and then choose c such that $\Delta + 4c$ is a square which we could then set to be b and since we have chosen a to be 1 the form we get will always be primitive. But we won't have the converse i.e we can't prove that C_Δ is a group $\forall \Delta$ since we have leave out some of the Δ due to our conditions.

More explicitly we have the following-

Proposition 4.1.2. *If Δ is the disc of $Q = ax^2 + bxy + cy^2$, such that $\Delta = \Delta_K$ i.e fundamental discriminant, then Q is primitive.*

Proof. If $\Delta = \Delta_K = 4s + 1$ and Q is non-primitive i.e $d \mid a, d \mid b, d \mid c$.

$$\Rightarrow d^2 \mid 4s + 1$$

but we know if $\Delta_K \equiv 1 \pmod{4}$ then $\Delta_K = d_1$ for some square free d_1 hence $d = 1$.

If $\Delta = \Delta_K = 16s + 8 = 4(4s + 2)$

if $d^2 \mid 4(4s + 2)$

$\Delta_K \equiv 0 \pmod{4}$ then $\Delta_K = 4d_1$, for some square free d_1 . if $d = 2$ then

$$\begin{aligned} (2b_1)^2 - 4 \cdot 4a_1c_1 &= 4(4s + 2) \\ \Rightarrow b_1^2 - 4a_1c_1 &= 4s + 2 \end{aligned}$$

Taking $\pmod{4}$ gives contradiction .

Similar technique for $\Delta_K = (16s + 12)$. □

Remark 4.1.3. The converse is not true, we have primitive forms which does not give field disc always i.e take $(4, 8, 1) \rightarrow \Delta = 48 \rightarrow$ which is not a field discriminant. So here we want to work with quadratic forms for all but specific forms (primitive) namely whose disc are of the $4s + 1, 16s + 8, 16s + 12$ as each of them represent discriminant of some quadratic field K . So

We set a quadratic number field $\mathbb{Q}(\sqrt{\Delta_K})$

Fix $\Delta = \Delta_K$ (Where Δ is the discriminant of the binary quadratic forms we would be dealing with).

We look at the set of all primitive forms with the given discriminant and denote them by $Cl(\Delta)$. Now under our $SL_2(\mathbb{Z})$ transformation we denote the equivalence classes by $C_\Delta = Cl(\Delta) / \sim$.

Chapter 5

Fractional Ideals and Ideal Class Group

Definition 5.0.1. Suppose D is an integral domain with the quotient field F . Then a non-empty subset I of F is called fractional - D ideal if it satisfies the following-

1. For any $\alpha, \beta \in I, \alpha + \beta \in I$
2. For any $x \in I$ & $r \in D, rx \in I$.
3. $\exists \gamma \in D$ such that $\gamma I \subseteq D$.

Proposition 5.0.1. $I^{-1} = \{\alpha \in F : \alpha I \subseteq O_K\} = \{O_K : I\}$ is a non-zero fractional O_K -ideal

Proof. Let $\alpha, \beta \in I^{-1}$ then

$$\begin{aligned}\alpha I &\subseteq O_K \text{ \& } \beta I \subseteq O_K \text{ so} \\ (\alpha + \beta)I &\subseteq \alpha I + \beta I \subseteq O_K.\end{aligned}$$

$\Rightarrow \alpha + \beta \in I^{-1}$. If $\alpha \in I^{-1}$ & $r \in O_K$ then $\alpha I \subseteq O_K$, then $r\alpha I \subseteq O_K$ from which it is clear that $r\alpha \in I^{-1}$.

Lastly let γ be a non-zero element of I , then for $\alpha \in I^{-1}$ $\alpha I \subseteq O_K$. So in particular $\alpha\gamma \in O_K$ hence $\gamma I^{-1} \subseteq O_K$.

Thus I^{-1} is a non-zero fractional O_K -ideal. $\square$

In an integral domain D , a fractional D -ideal I is called invertible if $I^{-1}I = D$. where I^{-1} is called the inverse of I .

To show every fractional O_K -ideal is invertible we need a lemma.

For this section we use O_K and A interchangeably.

Lemma 5.0.1. Every maximal ideal in A is invertible. Here K is the field of fractions of A .

Proof. Let m be a maximal ideal of A . Then $m \neq (0)$ since A is not a field

$$m' = \{x \in K \mid xm \subseteq A\} = (A : m).$$

Clearly m' is an A -submodule of K ; any non-zero element of m serves as a common denominator for elements of m' . Thus m' is a fractional ideal of A . It suffices to show that $mm' = A$.

$m'm = (A:m)$ $m \subseteq A$ is clear, on other hand

$A \subseteq m'$ so $m = Am \subset m'm$. As m is maximal and, $m \subset m'm \subset A$, either $m = mm'$ or $m'm = A$. It remains to show $m = mm'$ is not possible.

If $m = mm'$ and if $x \in m'$, then $xm \subset m$, $x^2m \subset m$ and thus by induction $x^n m \subset m$. Therefore for any $d \in m \exists \{d_i\}_{i \geq 1} \in m$ such that $x^i d = d_i$.

Thus any non-zero element $d \in m$ is a common denominator for all powers of x^n , $n \in \mathbb{N}$.

It follows $A[x]$ is a fractional ideal of A . Since A is Noetherian, $A[x]$ is finitely generated as A -module. (Since if A is a Noetherian domain then every fractional ideal is finitely generated as A -module.)

So x is integral over A , but A is integrally closed. Thus $x \in A \Rightarrow m'm = m \Rightarrow m' = A$.

It remains to show that it is impossible.

Let $a \neq 0$ and $a \in m$. The ideal Aa contains a product $p_1 p_2 \cdots p_n$. non-zero prime ideals. ($\because$ A be Noetherian domain $\Rightarrow$ every ideals contains a product of non zero prime ideals).

Taking n as small as possible, we have $m \supset Aa \supset p_1 \cdots p_n \Rightarrow m \supset p_i$ for some i . Say $i = 1$, As p_1 is maximal $\Rightarrow m = p_1$. Put $b = p_2 \cdots p_n$. Then. $Aa \supset mb$ and $Aa \not\subset b$, since n was the smallest. Thus $\exists b_1 \in b$ such that $b_1 \notin Aa$.

Since $mb \subset Aa \Rightarrow mba^{-1} \subset A \Rightarrow ba^{-1} \subset m'$, but $\because b \notin Aa, ba^{-1} \notin A$ (Contradiction since we assumed $A = m'$ and we showed an element belongs to m' but not in A). Hence $m' \neq A$. This concludes our proof. $\square$

Lemma 5.0.2. *Any fractional A ideal has the decomposition into prime ideals which is unique i.e there exists $p_i \in A$ such that for any fractional A -ideal I we have.*

$$I = \prod_{i=1}^n p_i^{\alpha_i} \text{ where } \alpha_i \in \mathbb{Z}$$

Since A is a Dedekind domain $\Rightarrow$ Prime ideals are maximal and since each maximal ideal is invertible so we can we can invert I .

Once we have this we can say.

- $I_A \rightarrow$ set of all fractional A ideal is an Abelian group.
- $P_A \rightarrow$ set of all principal fractional ideal is a subgroup of I_A .

Definition 5.0.2. $C_A = I_A/P_A$ Let P_A^+ denote the group of Principal fractional ideal (α) where $N(\alpha) > 0$,

$$I_A/P_A^+ = C_A^+$$

Now, $I \sim J$ if

$$IP_A = JP_A$$

$$\Rightarrow IJ^{-1} \in P_A \Rightarrow IJ^{-1} = (\alpha) \quad \alpha \in \mathbb{Q}(\sqrt{\Delta}) \text{ (the field of fraction of } A \text{)}$$

$$\Rightarrow (a)I = (b)J \text{ if } \alpha = a/b \text{ where } a, b \in A.$$

denote $\approx$ Strict equivalence for ideals in C_A^+ if

$$I \approx J \Rightarrow IJ^{-1} = (a) \quad a = \alpha^{-1}\beta \text{ \& } N(\alpha^{-1}\beta) > 0 \Rightarrow N(\alpha\beta) > 0$$

$$\Rightarrow (\alpha)I = (\beta)J, \alpha, \beta \in A$$

with $N(\alpha\beta) > 0$.

Proof. Since $\gamma I \subseteq A$ for some $\gamma \in A$. Thus $(\gamma)I = J$ where J is an A -ideal.

Thus we can prove the lemma for only A -ideals.

Existence:- Let S be the set of all non-zero proper ideal that are not so representable. if $S \neq \varphi$; then S has a maximal member M . ($\because A$ is Noetherian). By assumption M Can't be prime: and also $M \neq A$ also since $A = O_K$ is the product of empty collection of Prime ideals. So M is contained in some maximal ideal say I . Let I' be the inverse fractional ideal of I ;

$$\because M \subset I', MI' \subseteq II' = A$$

$$\text{And as } A \subset I', MA \subset MI' \Rightarrow M \subset MI'.$$

In fact $MI' \neq M$ (if $MI' = M$ and if $x \in I'$ then $xM \subset M \rightarrow x^n M \subset M \forall n$. Thus for same reasons as in [Lemma 5.0.1](#) we have x is integral over A and hence $x \in A$).

$$\Rightarrow I' \subset A \text{ But this is impossible } \because A \subset I'.$$

According to maximality of M in S we have $MI' \notin S$ & hence

$$\begin{aligned} MI' &= p_1 \cdots p_n. \\ \Rightarrow M &= Ip_1 \cdots p_n \end{aligned}$$

We have already proved that all maximal ideals are invertible and I is a maximal ideal. Thus we have a contradiction and hence S is empty.

Now for the **uniqueness** we can use the same logic as in the proof of **The Fundamental Theorem Of Arithmetic** [\[NZM\]](#) and hence our Lemma. $\square$

Chapter 6

Main Results

Now that we have developed many machinery let us use them to prove our main result. Here we would use C_A and C_{O_K} alternatively.

1. C_Δ has a group structure for all permissible Δ 's.
2. $|C_{O_K}|$ is finite, for all Quadratic Number Rings.

6.1 $|C_{O_K}|$ is finite for all Quadratic Number Rings

We would prove (1) by showing $\exists$ a bijection between $C_{O_K}^+$ and C_Δ . Since $C_{O_K}^+$ is a group (1) would follow. For now we assume this bijection and see how (2) follows.

Due to the bijection mentioned above we have $C_{O_K}^+ \cong C_\Delta$.

Clearly if $\Delta < 0$

$$C_{O_K} = C_{O_K}^+ \quad \because N(x) > 0 \quad \forall x \text{ \& hence } P_A^+ = P_A.$$

Thus $C_A \cong C_\Delta$ hence $|C_A| = |C_\Delta|$. This completes the case for imaginary quadratic fields.

When $\Delta > 0$.

if K is real quadratic field and K has a unit of norm -1 then-

Claim 6.1.1. $P_A^+ = P_A$.

Proof. Let $(\alpha) \in P_A$ such that $N(\alpha) < 0$. But then $N(u\alpha) > 0$ and $(u\alpha) = (\alpha)$ hence replace α with $u\alpha$, we are good to go.

Hence $P_A^+ = P_A$ □

$$\Rightarrow C_A^+ = \frac{I_\Delta}{P_\Delta^+} = \frac{I_\Delta}{P_\Delta} = C_A$$

Hence $C_A \cong C_\Delta$.

$$\Rightarrow |C_A| = |C_\Delta|.$$

if K does not have a unit of norm -1 then-

Claim 6.1.2. $[C_{O_K^+} : C_{O_K}] = 2$.

Proof.

Subclaim 6.1.1. $\alpha \sim -\alpha$ if and only if K has a unit of norm -1 .

Proof. If we have $N\left(p + \frac{\sqrt{\Delta}}{2}r\right) = -1$ then

$$\alpha \sim \begin{pmatrix} p & \frac{-\Delta r}{4} \\ r & -p \end{pmatrix} -\alpha$$

Here $\alpha = (1, 0, -\Delta/4)$.

Now if $\alpha \sim -\alpha$ by some $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$

then $p^2 - \frac{\Delta r^2}{4} = -1$.

Hence we have our required unit of norm -1 .

Similar result hold for $\alpha = (1, 1, (1 - \Delta)/4)$. □

Let H be the subgroup, of C_Δ such that $a > 0$,

$$\begin{aligned} \varphi : C_{O_K} &\rightarrow H \\ \left(a, \frac{-b + \sqrt{\Delta}}{2}\right) &\mapsto (a, b, c) \end{aligned}$$

See that this is an isomorphism (same proof as $C_{O_K^+} \cong C_\Delta$).

Hence $H \cong C_{O_K}$.

Subclaim 6.1.2. $[C_\Delta : H] = 2$.

Proof. Let $f \in C_\Delta$, f is arbitrary then f & $-f$ does not belong to same class in C_Δ/H .

If f & $-f$ belongs to same class in C_Δ/H then $f^{-1}(-f) \in H$.

Let $f = (a, b, c) \implies -f = (-a, -b, -c)$.

$$\begin{aligned} f^{-1} &\sim (c, b, a) & \text{but } \text{sign}(a) &= \text{sign}(c) \\ f^{-1}(-f) &\sim (-ac, *, *) & \implies ac &< 0 \end{aligned}$$

contradiction and hence they can't be in the same class.

(The sign of a and c are same as $f^{-1}f \sim (ac, b, 1) \sim (1, 0, \frac{\Delta}{4})$ since f and f^{-1} are always united forms and hence $ff^{-1} \in H$ hence f and f^{-1} are in same class and hence sign of a and c are same).

For arbitrary f & $g \in C_\Delta$ either (f, g) or $(f, -g)$ belong to same class in C_Δ/H . (Proof in same way as above).

So we get for any $f \in C_\Delta$ f and $-f$ does not belong to the same class in C_Δ/H and for any f & $g \in C_\Delta$ either f & g or f & $-g$ belong to same class in C_Δ/H .

Thus we get we have only 2 classes in C_Δ/H to choose from. hence $[C_\Delta : H] = 2$. □

$\because \alpha$ & $-\alpha$ doesn't belong to same class in C_Δ/H . Put them in different classes, then choose $f \in C_\Delta$ that does not belong to the class of α & put them in one of the class & so on. The process stops $\because |C_\Delta| < \infty$ for any Δ . [NZM]

Thus we have By lagrange Theorem

$$[C_{O_K}^+ : C_{O_K}] = \frac{|C_{O_K}^+|}{|C_{O_K}|} = \frac{|C_\Delta|}{|H|} = [C_\Delta : H] = 2$$

and hence our claim which completes our proof. $\square$

Thus $|C_{O_K}^+| = 2|C_{O_K}|$.

Thus we have-

$$|C_{O_K}^+| = |C_\Delta| = \begin{cases} |C_{O_K}|, & \text{if } \Delta < 0 \\ |C_{O_K}| & \text{if } \Delta > 0 \text{ \& } \exists u \in K \text{ with } N(u) = -1 \\ 2|C_{O_K}| & \text{if } \Delta > 0 \text{ \& } \nexists u \in K \text{ with } N(u) = -1. \end{cases}$$

Last 2 accounts of indefinite forms while the first one accounts for definite forms.

Now $\because |C_\Delta| < \infty \forall \Delta$ [NZM] and hence we get $|C_{O_K}| < \infty$. Hence ideal class number is finite for all Quadratic Number Rings.

6.2 C_Δ has a group structure for all Fundamental Discriminants Δ

Now we prove (1)

We will prove a few lemma first.

Lemma 6.2.1. *Let $[\alpha, \beta] = \alpha\mathbb{Z} + \beta\mathbb{Z}$ $[\gamma, \delta] = \gamma\mathbb{Z} + \delta\mathbb{Z}$ be 2 $\mathbb{Z}$ modules with $\alpha, \beta, \gamma, \delta \in R$ where $R = O_K$ $K = \mathbb{Q}(\sqrt{\Delta})$ then $[\alpha, \beta] = [\gamma, \delta]$ if and only if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = X \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$ where $X \in GL_2(\mathbb{Z})$.*

Proof. If $[\alpha, \beta] = [\gamma, \delta]$, then there are integers $x_0, y_0, z_0, \omega_0, x, y, z, \omega$ such that

$$\begin{aligned} \alpha &= x\gamma + y\delta, & \beta &= \omega\gamma + z\delta. \\ \gamma &= x_0\alpha + y_0\beta, & \delta &= \omega_0\alpha + z_0\beta \\ \Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} &= X \begin{pmatrix} \gamma \\ \delta \end{pmatrix}. \\ X &= \begin{pmatrix} x & y \\ \omega & z \end{pmatrix} \\ \begin{pmatrix} \gamma \\ \delta \end{pmatrix} &= \begin{pmatrix} x_0 & y_0 \\ \omega_0 & z_0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \\ \Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} &= XX_0 \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \\ \Rightarrow XX_0 &= I_2 \Rightarrow \det(X) \& \det(X_0) = \pm 1. \end{aligned}$$

Conversely if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = X \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$ holds then $[\alpha, \beta] \subseteq [\gamma, \delta]$ and $\because \det(X) = \pm 1$ we can multiply by its inverse to get (γ, δ) as linear combination of α & β hence $[\gamma, \delta] \subseteq [\alpha, \beta]$, and hence our lemma. $\square$

Lemma 6.2.2. *The form $f(x, y)$ properly represents n if and only if $f(x, y)$ is properly equivalent to the form $nx^2 + Bxy + Cy^2$ for some $B, C \in \mathbb{Z}$.*

Proof. If f is properly equivalent to the form $nx^2 + Bxy + Cy^2$ for some $B, C \in \mathbb{Z}$ then $\exists$

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SL_2(\mathbb{Z})$$

and hence $f(\alpha, \gamma) = n$ and $\gcd(\alpha, \gamma) = 1$.

Conversely if $\exists \alpha, \gamma$ such that $f(\alpha, \gamma) = n$ and $\gcd(\alpha, \gamma) = 1$, then by Bezout's Lemma $\exists \beta, \delta$ such that $\alpha\delta - \beta\gamma = 1$ and hence f is properly equivalent to the form $nx^2 + Bxy + Cy^2$ for some $B, C \in \mathbb{Z}$ by

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SL_2(\mathbb{Z})$$

which finishes our lemma. $\square$

Lemma 6.2.3. *Let $F = \mathbb{Q}(\sqrt{\Delta_F})$ be a quadratic field of discriminant Δ_F and let $m \in \mathbb{Z}$. Then every proper equivalence class of forms of discriminant Δ_F contains a primitive form with positive leading coefficient that is relatively prime to m .*

Proof. Let $f = (a, b, c) \in C_{\Delta_F}$ and set

$$P_{a,m,c} = \prod_p p$$

where the product ranges over all distinct primes p such that $p|a, p|c$ and $p \nmid m$. Also set

$$P_{a,m} = \prod_q q$$

where the product ranges over all distinct primes q such that $q|a, q|m$, but $q \nmid c$, set

$$P_{c,m} = \prod_r r$$

where the product ranges over all distinct primes r such that $r|c, r|m$, but $r \nmid a$, and set

$$S_m = \prod_s s$$

where the product ranges over all distinct primes s such that $s \mid m$ but $s \nmid P_{a,m,c}P_{a,m}P_{c,m}$. Then f represents

$$aP_{a,m}^2 + bP_{a,m}P_{c,m}S_m + c(P_{c,m}S_m)^2 = N \dots (*)$$

Claim 6.2.1. $\gcd(N, m) = 1$.

Proof. Assume that a prime $t \mid N$ and $t \mid m$. Assume first that $t \mid a$. Then

$$t \mid P_{a,m,c}P_{a,m}$$

by the definition of the latter. If $t \mid P_{a,m}$, then by (*),

$$t \mid cP_{c,m}S_m$$

However, $t \nmid P_{c,m}S_m$, so $t \mid c$. This contradicts the fact that $t \mid P_{a,m}$. Hence, $t \nmid P_{a,m}$, so $t \mid P_{a,m,c}$. It follows from (*) that

$$t \mid bP_{a,m}P_{c,m}S_m$$

However, we have already shown that $t \nmid P_{a,m}$ and since $t \mid a$, then $t \nmid P_{c,m}$. Also, $t \mid P_{a,m,c}$, so $t \nmid S_m$, which implies that $t \mid b$. We have shown that $t \mid \gcd(a, b, c)$, contradicting that f is primitive. Hence, our initial assumption was false, namely, we have shown that $t \nmid a$. Therefore,

$$t \mid P_{c,m}S_m$$

by the definition of the latter. However, by (*), this implies that $t \mid aP_{a,m}$, a contradiction to what we have already shown. This secures the claim. $\square$

By [Lemma 6.2.2](#), our claim tells us that f is properly equivalent to the form

$$g(x, y) = Nx^2 + Bxy + Cy^2$$

for some $B, C \in \mathbb{Z}$. If $N > 0$, then we have our result. If $N < 0$, then by setting $x_0 = Bm\ell + 1$ and $y_0 = -2N\ell m$ for some $\ell \in \mathbb{Z}$,

$$\begin{aligned} g(x_0, y_0) &= Nx_0^2 + Bx_0y_0 + Cy_0^2 \\ &= N(Bm\ell + 1)^2 + B(Bm\ell + 1)(-2N\ell m) + C(2N\ell m)^2 \\ &= NB^2m^2\ell^2 + 2NBm\ell + N - 2NB^2m^2\ell^2 - 2NB\ell m + 4CN^2\ell^2m^2 \\ &= N(1 - m^2\ell^2(B^2 - 4NC)) = N(1 - m^2\ell^2\Delta_F) = Q \end{aligned}$$

where $Q > 0$ if $N < 0$. Since f represents

$$Q = N(1 - m^2\ell^2\Delta_F)$$

and Q is relatively prime to m , given that N and $1 - m^2\ell^2\Delta_F$ are relatively prime to m , then [Lemma 6.2.2](#) gives us the complete result and finishes our lemma. $\square$

Now we prove

Theorem 6.2.1. *If C_Δ denotes the set of classes of primitive forms of fundamental discriminant Δ , then C_Δ forms a group with multiplication given by Dirichlet Composition and*

$$C_{O_K}^+ \cong C_\Delta$$

Proof. Let

$$\tau : C_{O_K}^+ \longrightarrow C_\Delta$$

$$\left\langle a, \frac{-b + \sqrt{\Delta}}{2} \right\rangle \mapsto (a, b, c) \quad \left[c = \frac{b^2 - \Delta}{4a} \right]$$

Now if

$$\begin{aligned} I &= \left\langle a, \frac{-b + \sqrt{\Delta}}{2} \right\rangle \approx J = \left\langle a', \frac{-b' + \sqrt{\Delta}}{2} \right\rangle \\ \Rightarrow (\alpha_1) \left\langle a, \frac{-b + \sqrt{\Delta}}{2} \right\rangle &= (\alpha_2) \left\langle a', \frac{-b' + \sqrt{\Delta}}{2} \right\rangle \quad [N(\alpha_1\alpha_2) > 0] \\ \Rightarrow (\alpha_1)(a^2a') \left\langle a, \frac{-b + \sqrt{\Delta}}{2} \right\rangle &= (\alpha_2)(a^2a') \left\langle a', \frac{-b' + \sqrt{\Delta}}{2} \right\rangle \\ \Rightarrow (\alpha_1a)(aa')I &= (\alpha_2a')(a^2)J \\ \Rightarrow (aa')I &\approx (a')J \quad [\because N(\alpha_1\alpha_2aa') > 0] \end{aligned}$$

as $N(aa') > 0$ as $a, a' \in \mathbb{N}$.

Thus if $I \approx J$ then without loss of generality we can assume $a = a'$.

We will denote members of $C_{O_K}^+$ by $\bar{I}$ & C_Δ by $\bar{f}$. Now if we have.

$$\begin{aligned} \bar{I}_f &= \bar{I}_{f_1} \dots (+) \\ \Rightarrow \tau(\bar{I}_f) &= \tau(\bar{I}_{f_1}) \quad \text{or } \bar{f} = \bar{f}_1 \quad \text{or } f \sim f_1 \end{aligned}$$

Since we have (+) ,by [Lemma 6.2.1](#) we get

$$\begin{pmatrix} a \\ \frac{-b_2+\sqrt{\Delta}}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ q & 1 \end{pmatrix} \begin{pmatrix} a \\ \frac{-b_1+\sqrt{\Delta}}{2} \end{pmatrix}$$

& hence

$$f \sim_{\begin{pmatrix} 1 & q \\ 0 & 1 \end{pmatrix}} f_1$$

Hence $\bar{f} = \bar{f}_1$ & hence τ is well defined.

Injectivity:

If $\tau(\bar{I}) = \tau(\bar{J}) \Rightarrow \bar{I} = \bar{J}$

Let

$$\begin{aligned} \bar{I} &= \left\langle a, \frac{-b + \sqrt{\Delta}}{2} \right\rangle \mapsto f = (a, b, c) \\ \bar{J} &= \left\langle a', \frac{-b' + \sqrt{\Delta}}{2} \right\rangle \mapsto g = (a', b', c'). \end{aligned}$$

Again by same logic as can assume.

$$a = a'$$

Now Comparing Coefficients we get

$$b = b' \quad \& \quad c = c'$$

Thus $\bar{I} = \bar{J}$. Thus τ is injective.

Surjectivity:

By [Lemma 6.2.3](#) we have every proper equivalence classes of forms of discriminant Δ contains a primitive form with positive coefficient relatively prime to m . Just choose those forms f_i from each of the classes and map them to the class containing the ideals $\bar{I}_{f_i}$ and hence τ is surjective.

Now

Claim 6.2.2. τ respects Dirichlet composition and satisfies

$$\tau(\bar{I}\bar{J}) = \tau(\bar{I}) \circ \tau(\bar{J})$$

Proof. Easy to see by our previous calculations that this holds if $\tau(\bar{I})$ and $\tau(\bar{J})$ are united. So we look at the case when they are not united forms.

We know $\exists I_1 \sim I$ and $J_1 \sim J$ such that $\exists \bar{f} \ \& \ \bar{g} \in C_\Delta$ [due to surjectivity] such that $\tau(\bar{I}_1) = \bar{f}$ and $\tau(\bar{J}_1) = \bar{g}$.

$\therefore f$ well defined

$$\tau(\bar{I}_1) = \tau(\bar{I}), \quad \tau(\bar{J}_1) = \tau(\bar{J}).$$

$$\& \quad \bar{I}_1 \bar{J}_1 = \bar{I} \bar{J} \Rightarrow \tau(\bar{I}_1 \bar{J}_1) = \tau(\bar{I} \bar{J})$$

Thus we may as well work with $\tau(\bar{I}_1)$ & $\tau(\bar{J}_1)$.

Let $\tau(\bar{I}_1) = (a, b, c)$.

$\tau(\bar{J}_1) = (a_1, b_1, c_1)$.

By [Lemma 6.2.3](#) we know we can choose $\bar{f}$ and $\bar{g}$ such that $\gcd(a, a_1) = 1$ and hence $\bar{f}$ and $\bar{g}$ are united and hence our Dirichlet composition is defined.

Thus $\tau(\bar{I}_1) \circ \tau(\bar{J}_1)$ is defined. and since by our previous calculations we have shown that $\tau(\bar{I}_1 \bar{J}_1) = \tau(\bar{I}_1) \circ \tau(\bar{J}_1)$ holds for primitive forms thus we have-

$$\tau(\bar{I}\bar{J}) = \tau(\bar{I}_1 \bar{J}_1) = \tau(\bar{I}_1) \circ \tau(\bar{J}_1) = \tau(\bar{I}) \circ \tau(\bar{J}) \quad [\because \tau(\bar{I}_1) = \tau(\bar{I}), \quad \tau(\bar{J}_1) = \tau(\bar{J})]$$

This proves our claim. □

$$(ac, b, 1) \sim \begin{pmatrix} \frac{b}{2} & -1 \\ -1 & 0 \end{pmatrix} (1, 0, \Delta/4). \text{ When } \Delta = 0(\text{mod } 4).$$

$$(ac, b, 1) \sim \begin{pmatrix} \frac{-(b+1)}{2} & -1 \\ -1 & 0 \end{pmatrix} (1, 0, (1 - \Delta)/4). \text{ When } \Delta = 1(\text{mod } 4).$$

and also if $f = (a, b, c)$

$(a, b, c) \circ (1, 0, \Delta/4) = (a, b, c)$ and respectively $(1, 1, (1 - \Delta)/4)$

and

$$(a, b, c) \circ (c, b, a) = (ac, b, 1) \sim (1, 0, 4/4) \text{ or } (1, 1, (1 - \Delta)/4).$$

The claim above along with the observations above proves that C_Δ is forms a group for all the permissible Δ 's &

$$C_\Delta \cong C_{O_K}^+$$

□

Remark 6.2.1. We can give a group structure to $C_\Delta \forall \Delta$ using Gauss composition. [\[Flath\]](#)

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